

Algebraic expressions & formulae

Expansion

1. $(a + b)^2 = a^2 + 2ab + b^2$

2. $(a - b)^2 = a^2 - 2ab + b^2$

3. $(a + b)(a - b) = a^2 - b^2$

11. Expand $(2a - 3b^2 + 2)(a + b)$.

12. Simplify $-8(3a - 7) + 5(2a + 3)$.

Solve each of the following equations.

(a) $2(x - 3) + 5(x - 2) = 19$

(b) $\frac{2y+6}{9} - \frac{5y}{12} = 3$

Factorisation

↳ extracting common factors

⇒ An algebraic expression may be factorised by extracting common factors,
e.g. $6a^3b - 2a^2b + 8ab = 2ab(3a^2 - a + 4)$

⇒ An algebraic expression may be factorised by grouping,
e.g. $6a + 15ab - 10b - 9a^2 = 6a - 9a^2 + 15ab - 10b$
 $= 3a(2 - 3a) + 5b(3a - 2)$
 $= 3a(2 - 3a) - 5b(2 - 3a)$
 $= (2 - 3a)(3a - 5b)$

An algebraic expression may be factorised by using the formula $a^2 - b^2 = (a + b)(a - b)$,

e.g. $81p^4 - 16 = (9p^2)^2 - 4^2$
 $= (9p^2 + 4)(9p^2 - 4)$
 $= (9p^2 + 4)(3p + 2)(3p - 2)$

An algebraic expression may be factorised by inspection,

e.g. $2x^2 - 7x - 15 = (2x + 3)(x - 5)$

2x	3	3x
x	-5	-10x
2x ²	-15	-7x

Q4. Solve the equation $3x^2 - 2x - 8 = 0$.

Q Solve the equation $2x^2 + 5x - 12 = 0$.

Q 6. Solve $2x + x = \frac{12+x}{x}$.

Fractions - Addition & subtraction

⇒ To add or subtract algebraic fractions, we have to convert all the denominators to a common denominator.

Q7. Express each of the following as a single fraction.

(a) $\frac{x}{3} + \frac{y}{5}$

(b) $\frac{3}{ab^3} + \frac{5}{a^2b}$

(c) $\frac{3}{x-y} + \frac{5}{y-x}$

(d) $\frac{6}{x^2-9} + \frac{3}{x-3}$

Q8- Solve $\frac{4}{3b-6} + \frac{5}{4b-8} = 2$.

Fraction - multiplication & division.

⇒ To multiply algebraic fractions, we have to factorise the expression before cancelling the common terms. To divide algebraic fractions, we have to invert the divisor and change the sign from $\div$ to $\times$.

✓ keep, change, flip

Q 9. Simplify.

(a) $\frac{x+y}{3x-3y} \times \frac{2x-2y}{5x+5y}$

(b) $\frac{6p^3}{7qr} \div \frac{12p}{21q^2}$

(c) $\frac{x+y}{2x-y} \div \frac{2x+2y}{4x-2y}$

Changing the subject of a formula

→ subject, variables

Q10. Make t the subject of the formula, $v = u + at$.

Q11. Given that $T = 2\pi\sqrt{\frac{L}{g}}$, express L in terms of π , T and g .

It is given that $p = \frac{u^2 - v^2}{q}$.

- (a) Find p when $q = 3$, $u = -5$ and $v = 7$.

Answer $p = \dots\dots\dots$ [1]

- (b) Express v in terms of p , q and u .

Answer $v = \dots\dots\dots$ [2]

(a) Factorise $4x^2 - 9$.

Answer [1]

(b) Hence, express $\frac{1}{2x+3} - \frac{x}{(4x^2-9)}$ as a single fraction in its simplest form.

Answer [2]

Expand and simplify the following expressions.

(a) $-ab(7a - b)$

Answer [1]

(b) $(2p + 3q)(p - q)$

Answer [1]

Factorise the following expressions completely.

(a) $21p^2 + 15p$

Answer [1]

(b) $4ax - 21by + 6ay - 14bx$

Answer [2]